

Microbiology: Zoonotic Infections

Tony A. Slieman, M.Sc. Ph.D.

Director of Academic Affairs and Associate Professor

Founding Director of Masters in Biomedical Sciences

Department of Biomedical and Anatomical Sciences

tslieman@nyit.edu



Do.
Make.
Heal.

Innovate.

Reinvent the Future.

Session Objectives

1. Explain how epidemiology, microbial virulence factors, and host-defense mechanisms contribute to the prevention and development of zoonotic infections affecting the skin, blood, lymphatic systems, and other sites.
2. Differentiate among key zoonotic pathogens based on their unique microbiologic characteristics and their involvement in infections of the skin, blood, lymphatic systems, and other anatomical sites.
3. Apply common clinical microbiology testing procedures to identify major pathogens responsible for zoonotic infections of the skin, blood, and lymphatic systems.
4. Interpret microbiological results to support accurate clinical diagnosis of zoonotic diseases.

Pathogens/Microorganisms Covered

1. *Yersinia*
2. *Bacillus anthracis*
3. *Borrelia*
4. *Leptospira*
5. *Rickettsia*
6. *Anaplasma* and *Ehrlichia*
7. *Coxiella*
8. *Francisella*
9. *Brucella*
10. Other: Zoonotic influenza, Salmonellosis, West Nile, SARS, Lyme Disease

ZOONOTIC INFECTIONS

1. Infection acquired from any type of animal:
 - a. Vertebrates (mammals; fishes, etc.)
 - b. Invertebrates (insects / arthropods)
2. Often acquired from pets
3. Children are usually the most affected group
4. Occupational exposure (e.g. veterinarians; landscapers, national parks workers, and people who work in meat-packing plants)

Route of exposure

1. Direct skin penetration

- **Saliva** – animal bites [rabies: dogs & bats]
- Animal **scratches** [cat scratch disease]
- **Urine** from dogs & rats [leptospirosis]
- **Insect bites** [RMSF; Lyme disease; malaria]



Common Routes of exposure

2. Inhalation (from aerosolized microbes)

- Anthrax: from spores
- Q fever [mostly from farm animals]

3. Ingestion

- Cryptosporidiosis [contaminated water]
- Leptospirosis [contaminated water / food]
- Anthrax [contaminated food]

Yersinia

- Gram-neg, facultative anaerobic rod
- Encapsulated: LPS
- *Y. pestis* → bubonic plague
- *Y. enterocolitica* → enterocolitis, pseudo-appendicitis
- Cold resistance, enhanced isolation at 4°C
- Pathogenesis + Immunity:
 - Resistance to phagocytosis via type III secretion system (YopH gene product); ↓cytokine production, inflammation
 - Capsule + plasminogen activator protease degrade C4b and C5a → prevents opsonization

Yersinia - Epidemiology

- *Y. pestis*:
 - Sylvatic plague (rodents, cats) → urban plague (dogs)
 - Pandemic in 1320s killed 25 million in Europe
 - Modern: <10 cases/year in USA
 - Human : Human spread via pulmonary involvement
- *Y. enterocolitica*:
 - 1/100,000 people infected per year in U.S. (meat, milk, water)

Yersinia – Clinical Diseases

- *Y. pestis*:
 - Bubonic: Infected flea bite → fever and bubo (axillary/inguinal lymphatic swelling) 7 days later → 75% death rate w/o treatment
 - Pneumonic: 2-3 day incubation, fever and pulmonary symptoms w/n 1 day; 90% death rate w/o treatment
- *Y. enterocolitica*: Enterocolitis + Pseudoappendicitis (kids)
 - Avg. incubation = 4-6 days → bloody diarrhea, fever, abdominal pain x 1-2 weeks
 - Terminal ileum, mesenteric lymph nodes
 - Other: septicemia, intra-abdominal abscess, arthritis, hepatitis, osteomyelitis

Yersinia – Lab Diagnosis

- Isolation: stool, blood, bile, wound, throat swab, mesenteric lymph node, CSF, peritoneal fluid
- Labs use cefsulodin-irgasan-novobiocin (CIN) culture medium
- *"Fatal Laboratory-Acquired Infection with an Attenuated Yersinia pestis Strain – Chicago, IL, 2009, MMWR, 60(07);201–205 (Feb.25, 2011)".*

Bacillus anthracis

- Gram + rod, obligate spore-forming aerobe, encapsulated, lethal cause of anthrax; bioterrorism
- Physiology/Structure:
 - Long chains of single/double rods
 - Three toxins on one plasmid
 - Protective Antigen (PA)
 - Edema Factor (EF): CaM-dependent AC $\rightarrow \uparrow$ cAMP $\rightarrow$ edema
 - Lethal Factor (LF): Zn metalloprotease
- LF $\rightarrow$ macrophage apoptosis
- LF + EF $\rightarrow$ inhibition of innate immunity
- Capsule: poly D-glutamic acid

PA + EF =
Edema Toxin

B. anthracis: Epidemiology

- Unvaccinated herbivores (livestock) → accidental human infection
- 2001: USPS purposeful contamination (spore-induced infection of a postal employee)
- Route: inoculation/ingestion/inhalation
- Inhalation = “wool-sorter’s disease”
- Spores are highly resilient
- No person-person transmission

B. anthracis – Clinical Diseases

- Cutaneous Anthrax
 - Painless papule → ulcer w/surrounding vesicles → necrotic eschar (+ painful edema and lymphadenopathy [LAD])
 - *Necrosis is due to Lethal Factor (LF)*
- GI Anthrax:
 - Upper: regional LAD, sepsis, edema
 - Lower: Nausea, vomiting, malaise
- Inhalation Anthrax: alveolar macrophages deliver spores to mediastinal LN's (latency period) → nonspecific URI → mediastinal LAD, fever, massive edema → shock + death within 3 days



"Cutaneous Anthrax" N Engl J Med 2009; 361:178
DOI: 10.1056/NEJMicm0802093

B. anthracis – Lab + Clinical Diagnosis

- Visible on Gram stain: long, thin rods, Gram+, chains
- Low-CO₂ culture can grow spores
- Difficult distinction from *B. cereus* (*B. anthracis* is non-hemolytic, nonmotile)
- India ink exposes capsule
- “medusa head” colonies on sheep blood agar

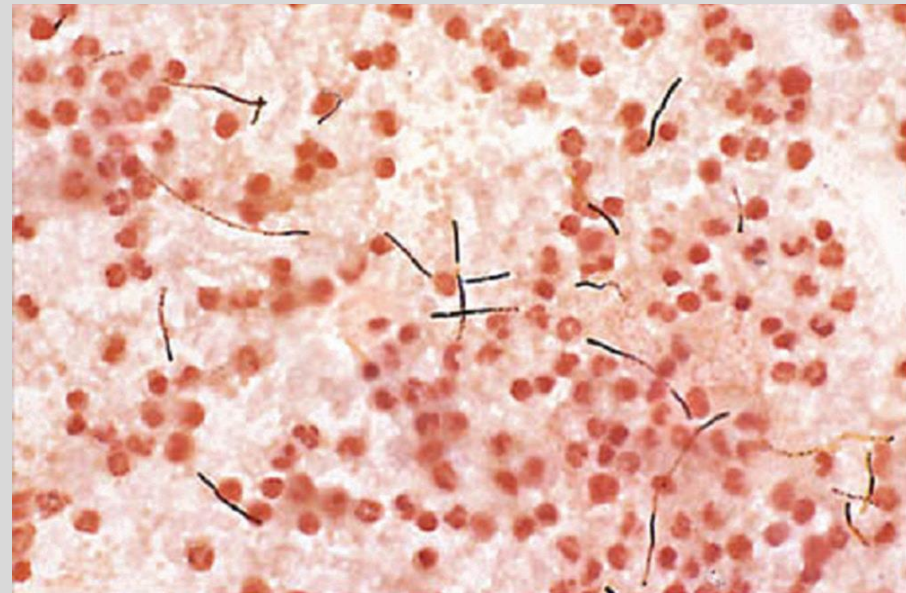


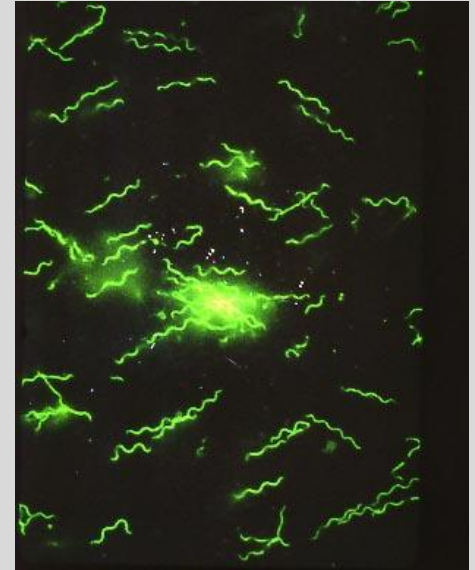
Fig. 20-2 (Murray): *B. anthracis* in blood from patient with pulmonary anthrax

B. anthracis – Treatment, Prevention, Control

- Empiric treatment = Ciprofloxacin or Doxycycline + 1-2 additional (Rifampin, Vancomycin, Penicillin, Imipenem, Clindamycin, Clarithromycin)
- Amoxicillin for cutaneous anthrax
- Vaccination: animals themselves, residents of endemic regions, handlers of animal products from endemic regions, and certain military personnel.

Borrelia

- Gram –ve spirochete (microscopy)
- *B. burgdorferi* & *B. recurrentis*
- Transmitted by *Ixodes* ticks
- Diseases: (1) Lyme disease (caused by *B. burgdorferi*) & (2) relapsing fever, due to *B. recurrentis*
- Poor Gram staining → stain with Giemsa or Wright



B. burgdorferi on fluorescence microscopy

Borrelia burgdorferi

- Transmission: The bacteria are transferred to humans via *Ixodes* saliva after a bite
- Immunity: Antibodies develop over weeks/months – can be maintained for years

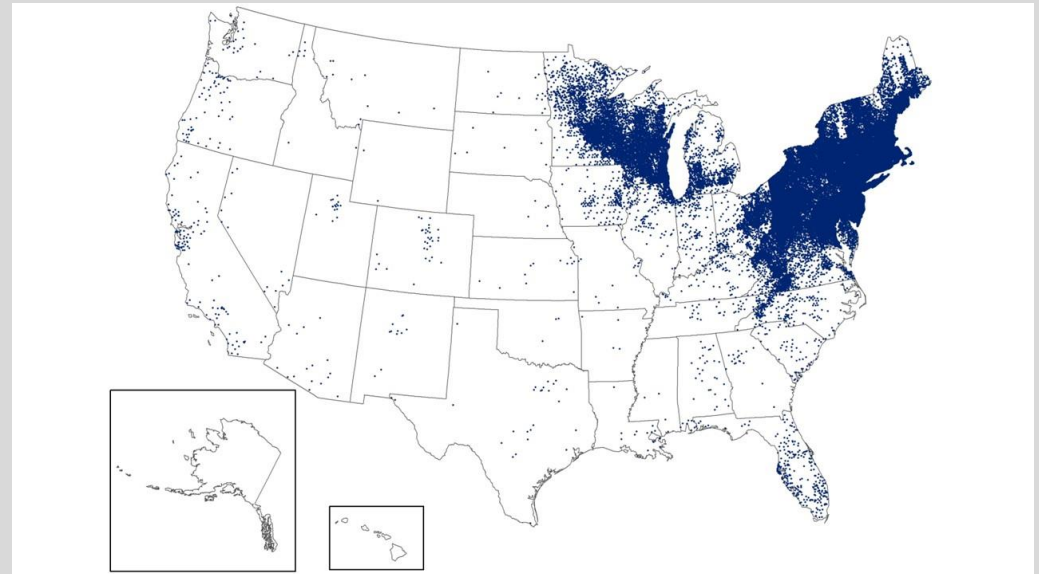


<https://entomologytoday.org/2023/01/13/single-mow-park-trails-not-reduce-ticks/blacklegged-tick-ixodes-scapularis-4/>

Lyme Disease- Epidemiology

- Leading vector-borne disease in U.S., 95% from Maine to Virginia and Upper Midwest
- Reservoir is white-footed mouse
- White-tailed deer essential for tick life cycle

“Reported Cases of
Lyme Disease —
United States, 2023”



Lyme Disease: Clinical Manifestations

- Erythema migrans rash ~5 cm diameter or ≥ 1 late manifestation (MSK, nervous system, cardiovascular) + lab confirmation
- Lab definition:
 - Isolation of *B. burgdorferi* [not usually done]
 - Dx levels of IgM or IgG Ab's
 - Significant increase in Ab titer between acute/convalescent stages
- Treatment: doxycycline



Lyme Disease: Clinical Stages

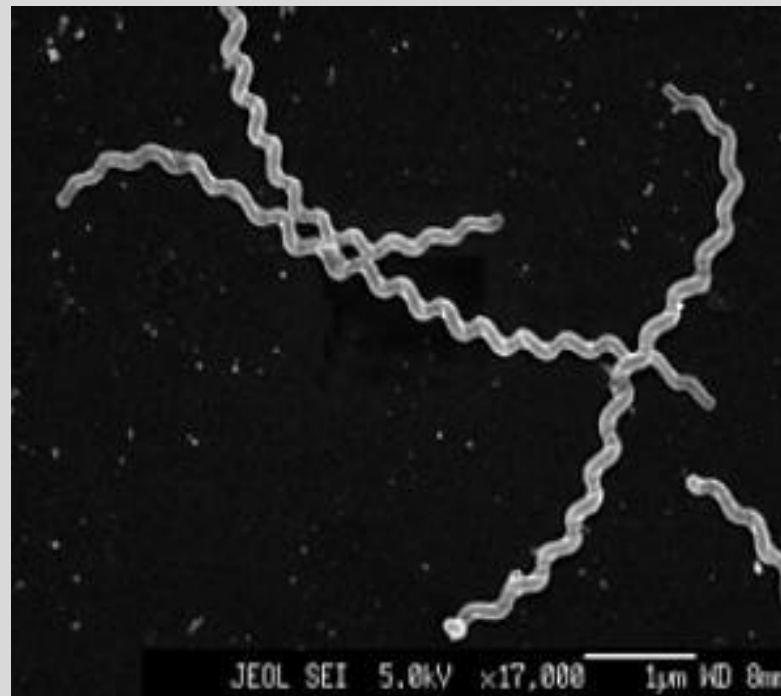
Stage	Timeframe	Features
1	1-30 days	Erythema migrans (chest, axilla), flulike symptoms, myalgia, arthralgia, HA, neck stiffness
2	3-10 weeks	Migratory polyarthrititis (knee, ankle, wrist), can recur, Bell's palsy, meningeal symptoms, <i>Borrelia</i> encephalopathy
3	Months-years	Lyme arthritis (LG joints), subacute encephalopathy, acrodermatitis chronic atrophicans

Relapsing Fever- Epidemiology

- *Borrelia recurrentis*; vector human body louse
- Requires war zone/crowded area for high lice load
- Modern disease burden: Ethiopia, Eritrea, Somalia, Sudan
- Antigenic variation → an important virulence factor which allows for its disease manifestations to be intermittent/continuous despite production of antibodies.
- High mortality if not treated (~70%)

Leptospira – Physiology, Structure

- Thin spirochetes, hooked ends, flagellated, obligate aerobes, require supplemented media



Leptospira – Pathogenesis & Immunity

- Subclinical infection
- Mild flu-like illness
- Systemic disease (Weil Disease)
- Penetration of mucous membranes, cuts
- Location:
 - Early: blood, CSF
 - Late: urine
- Some clinical manifestations from immunologic reactions

Leptospira - Epidemiology

- 100-200 U.S. infections per year (>50% in Hawaii)
- Reservoir hosts: hosts (colonize renal tubules → excreted in urine → animal urine (esp. rats) infects water sources)
- Incidental hosts: dogs, livestock, humans
- Warm months, recreational water exposure



Leptospira: Clinical Disease

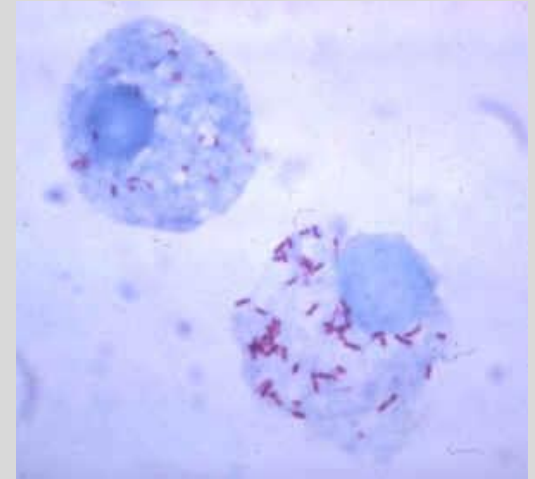
- Infection through skin or conjunctiva
- 1-2 week incubation → flu-like illness (fever, myalgia, chills, HA, vomiting, diarrhea) →
 - remission or
 - more severe illness: HA, myalgia, chills, abdominal pain, conjunctival suffusion →
 - Severe disease = vascular collapse, TCP, hemorrhage, hepatic/renal dysfunction
- Weil Disease = severe, icteric disease
 - No hepatic necrosis, usually full renal recovery

Leptospira

- Slow, specific culture (Fletcher, EMJH, Tween 80-albumin)
- Darkfield microscopy only if cultured prior
- Antibody detection favored over culture (microscopic agglutination test [MAT])
- Treatment: IV penicillin or doxycycline
- Prophylaxis: doxycycline only

Rickettsia rickettsii

- Rocky Mountain Spotted Fever
- Obligate intracellular Gram - rod (doesn't stain well)
- Stains best with Giemsa or Gimenez stain
- Arthropod vectors, rodent reservoirs
 - *Dermacentor* (dog tick)
- Transovarian transmission permits ticks to be hosts and vectors

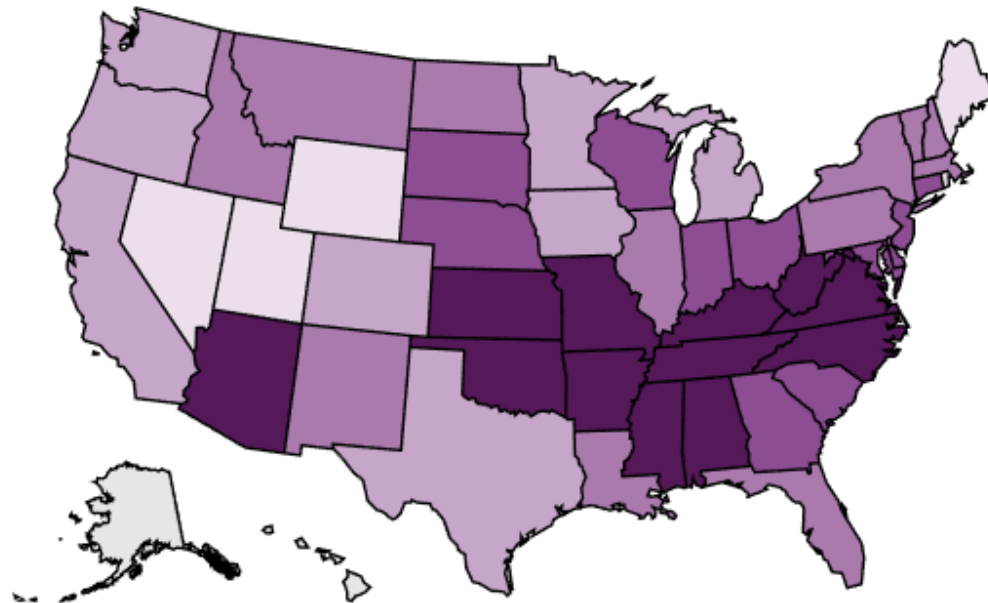


Rickettsia growing inside cultured cells: appear as short rods using a special stain

Rickettsia rickettsii

Annual incidence (per million population) of reported spotted fever rickettsiosis—United States for 2023

● 0 ● 0 to < 0.71 ● 0.71 to < 2.04 ● 2.04 to < 4.85 ● 4.85+ ● Not Notifiable



Arkansas, Missouri, North Carolina, Tennessee & Virginia
account for over 50% of cases

Rickettsia rickettsii

- Endothelial adherence via OmpA (Virulence Factor)
- Vascular leakage → hypovolemia, hypoproteinemia
- Cytokine-mediated killing via CD8+ T-cells
- Tick bite → macular rash (wrists, arms, ankles → trunk) 3 days post-tick bite
 - 7 days post-tick bite: high fever, HA, malaise, myalgia, N/V/Abdominal Pain, diarrhea
- Sequelae:
 - rash turns petechial (poor prognosis)
 - Renal and pulmonary failure, neurologic complications, cardiac abnormalities
 - 10-25% fatality rate if untreated
 - Endotoxin related



Rash of RMSF in early (top) and late (bottom) stages.

Image: CDC



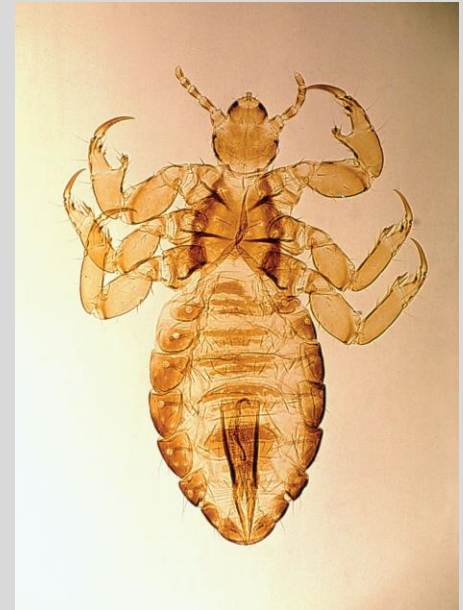
R. rickettsii: Diagnosis, Treatment

- Weil-Felix test (agglutination): insensitive, nonspecific; ↓cost, used in ↓resource settings
- Microimmunofluorescence more sensitive and specific
- Doxycycline



Rickettsia prowazekii

- Epidemic typhus
- Human to human transmission by human louse (*Pediculus humanus*)
- Central + South America
- Recrudescent disease (Brill-Zinsser disease) possible after first infection
 - Affects military personnel



<https://www.britannica.com/animal/human-louse>



Anaplasma and *Ehrlichia*

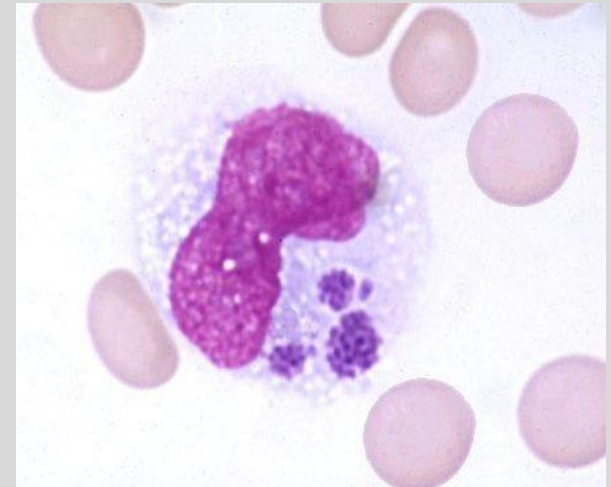
- Intracellular: mostly granulocytes (PMNs-*Anaplasma*), or monocytes – *Ehrlichia chafeensis*)
- Prevent phagolysosome fusion
- Gram –ve but no PG/LPS
- Giemsa, Wright staining
- Cross reaction common on serology
- Tick transmission, vertebrate hosts

Genus	Vector	Host
<i>Ehrlichia</i>	Lone Star Tick (<i>A. americanum</i>)	White-tailed deer
<i>Anaplasma</i>	<i>Ixodes</i>	Small rodents



Human Monocytic Ehrlichiosis

- *Ehrlichia chafeensis*
- Emerging disease (1986) in US
- Tick bite → 1-2 weeks later: high fever, HA, malaise, myalgia → late-onset rash
 - Most patients develop leukopenia, thrombocytopenia, elevated transaminases
 - Immunocompromised patients can develop sepsis
- Most patients require hospitalization (2-3% mortality)
- Reservoir white-tailed deer



Infected monocyte with 3
morulae

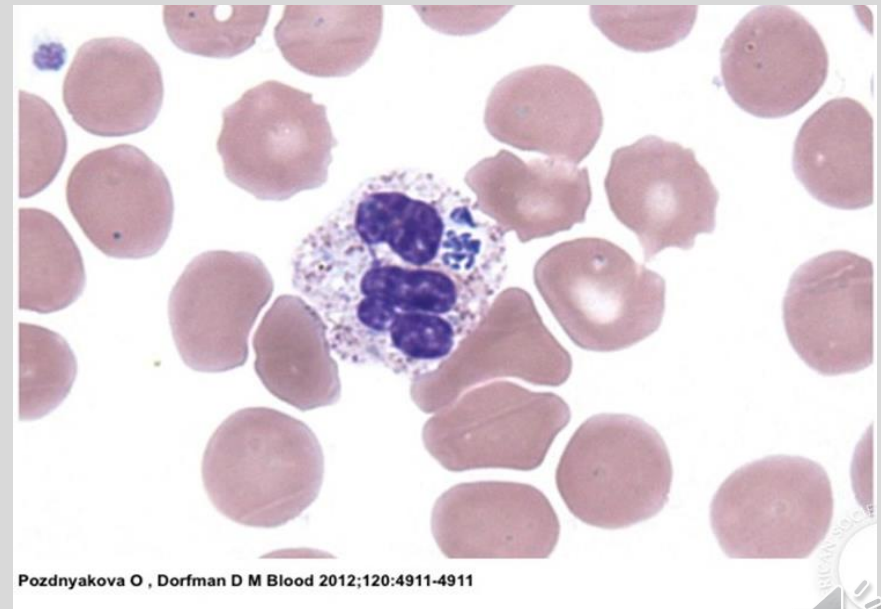


Anaplasmosis

- *Anaplasma phagocytophilum*
- Mostly granulocytes (PMNs)
- Tick bite → 5-10 days later: high fever, HA, malaise, myalgia
 - TCP, leukopenia, ↑transaminases (same as Ehrlichiosis)
- Sequelae: peripheral neuropathy

A neutrophil infected
with *Anaplasma*
phagocytophilum

Blood 2012;
120:4911-4911



Ehrlichiosis + Anaplasmosis: Diagnosis & Treatment

- Giemsa stained preps only useful in first week of infection; Gram stain ineffective
- Nucleic acid amplification and serology
 - Titers increase 3-6 weeks after presentation
 - Serology cross-reaction: RMSF, Q fever, Lyme, Brucellosis, EBV
- Treatment: doxycycline



Coxiella burnetii

- Gram -ve, weak Gram stain
- Q-fever
- *Coxiella* delays apoptosis
- Primary reservoirs: sheep, goats, cattle
- Sources: aerosols of tick feces, placentas, feces + urine of infected animals
 - Look for veterinarians, food handlers, ranchers, & farm personnel



Coxiella

- Mild symptoms → abrupt high-grade fever, fatigue, HA, myalgia
 - Severe sequelae (~5%): hepatitis, pneumonia, fevers
 - Look for granulomas in involved organs
 - Chronic: usually subacute endocarditis if immunosuppressed/preexisting valvulopathy



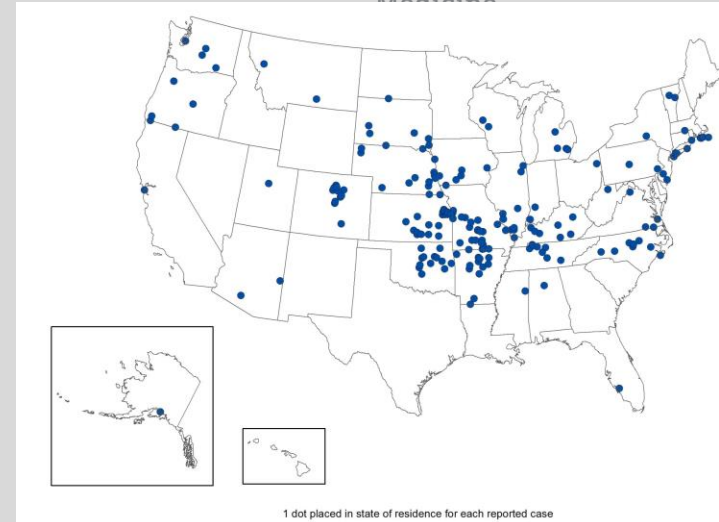
Coxiella: Diagnosis, Treatment

- Most common: serology (immunofluorescence antibody)
- Treatment: doxycycline for 14 days
 - Chronic: doxy. + hydroxychloroquine
- Vaccine: inactivated, whole-cell organisms
 - Uninfected animals, animal workers; no booster

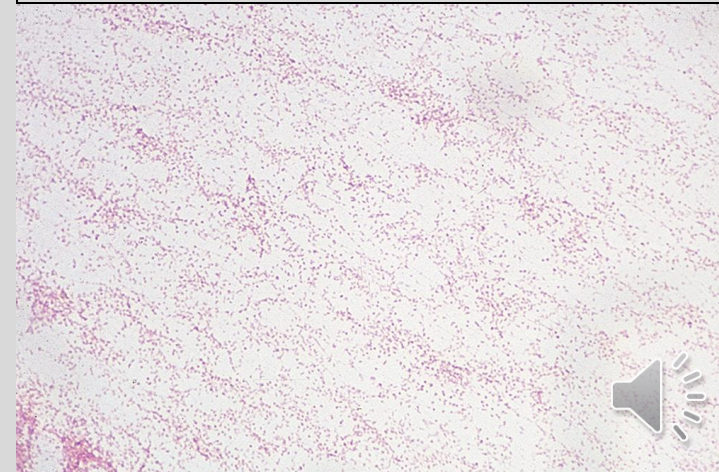


Francisella

- Multiple species in genus, most important is *F. tularensis*
- Aerobic Gram -ve, faintly staining coccobacillus, intracellular
- Culture: fastidious (cysteine; >3 days for growth)
- Culture hazardous
- Capsule
- Transmitted by ticks & deer flies; exposure to rabbits



Above: 2023 Infections (CDC)
Below: *F. tularensis* Gram stain (Murray
Fig. 29-4)



Tularemia

- AKA Glandular Fever/Rabbit Fever/Tick Fever/Deerfly Fever
- Type A infections: rabbits, hares, ticks (*Ixodes*, *Dermacentor*, *Amblyomma*)
- Most common manifestation: ulceroglandular (papule → necrotic ulcer with lymphadenopathy and bacteremia)
- Other types: oculoglandular, pneumonic, typhoidal, oropharyngeal, gastrointestinal
- In US: primarily in OK, AR and MO



Glandular Tularemia
(NEJM)



F. tularensis: Lab Diagnosis + Treatment

- Extreme biohazard in specimen collection
- Growth on buffered charcoal yeast extract (BCYE) agar w/cysteine supplement
 - BCYE also used for *Legionella*
- Ab titers usually used for Dx
- Treatment of choice: Gentamicin
 - Alternate: Doxycycline, Ciprofloxacin
- Prompt tick removal can prevent infection



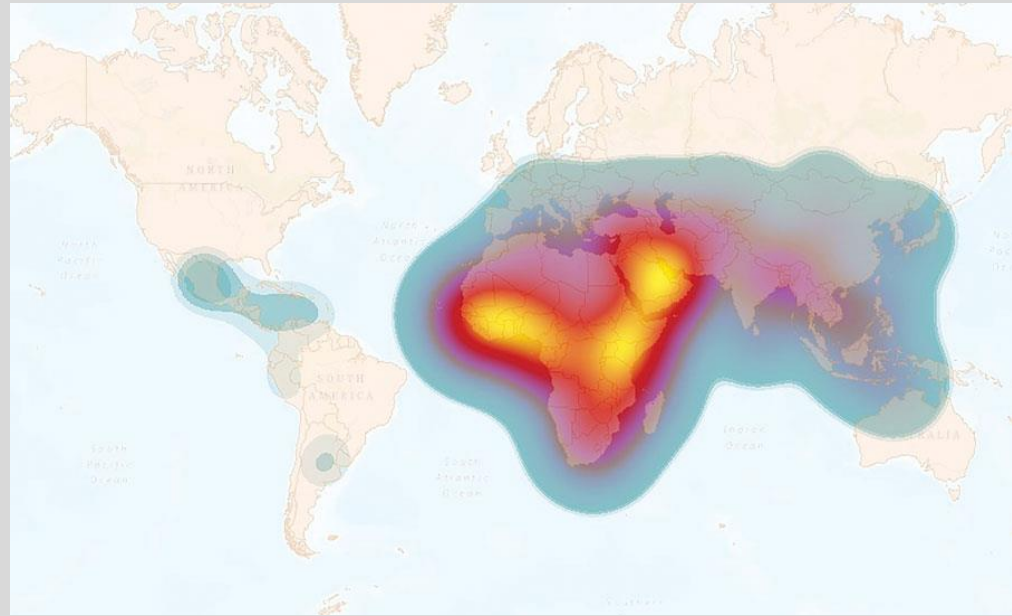
Brucella

- Four major species cause Brucellosis (aka “undulant fever”):
 - *B. abortus*; *B. melitensis*; *B. canis*; *B. suis*
- Non-motile, non-encapsulated, Gram -ve, coccobacillus
- Aerobic, no fermentation, slow growing
- Reticuloendothelial intracellular pathogen
 - Sequestration: spleen, liver, kidney, bone marrow, lymph nodes



Brucella Epidemiology

- Endemic: Latin America, Africa, Mediterranean Basin, Middle East, West Asia
- U.S.: annually between 80-140 cases
- Cattle, pigs, sheep, and goats
- Route: dairy products, inhalation



Global Heat map (2023) <https://wwwnc.cdc.gov/eid/article/29/9/23-0052-f2>



Brucella Lab Diagnosis

- ≥ 4 -fold titer increase from acute to convalescent phase of illness (≥ 2 weeks apart)
- Culture on blood agar or other enrichment media x 2 weeks before determining negativity
- Presumptive Dx: 4-fold increase or any single titer higher than 1:160



Brucella Clinical Syndromes

- Localization to organs with ↑erythritol
 - Breast, uterus, placenta, epididymis
- Symptoms 1-3 weeks after exposure: malaise, chills, sweat, fatigue, weight loss, nonproductive cough
- Fever can wax and wane: undulant fever
- Inadequate Tx produces chronic infection 3-6 mos. after antibiotics stopped:
 - Bone (osteomyelitis), spleen, liver are other sites of infection



Brucella Treatment

- Doxycycline + Rifampin
- Pregnant women and children ≤ 8 years old: trimethoprim-sulfamethoxazole instead of doxy.
- Treat for ≥ 6 weeks
- Prevention: animal vaccination (live attenuated organisms), avoidance of unpasteurized dairy products



Case 1

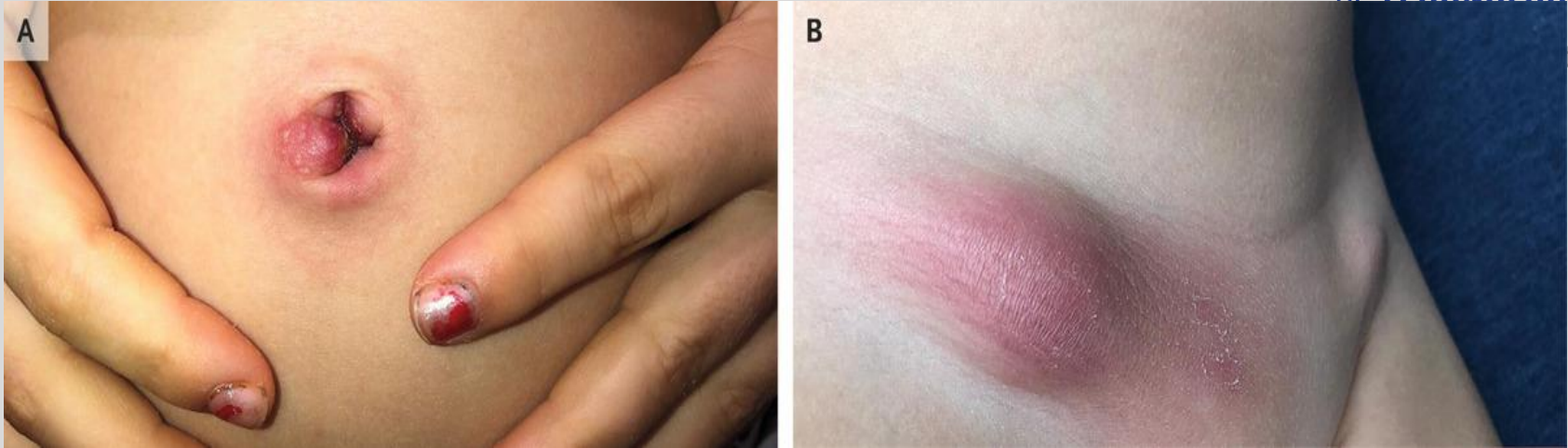
A 35-year-old male presents to his family physician with a 2 week history of joint pain in his left knee area. He recalls having a low-grade fever and malaise 4 weeks earlier. He works as a landscaper on Eastern Long Island. Upon questioning, he reveals that he had a rash about a month earlier, and he shows the physician a photo of the rash (shown below):



- What is the likely pathogen and pathogenesis of this disease?
- How does this disease evolve clinically?



Tularemia cont'd



A 5-year-old girl presented to the pediatric emergency department with a 4-week history of painful swelling on both sides of her lower abdomen. Pets that she had regular contact with included a cat and a dog. Six weeks before presentation, her parents had noticed a tick buried in her umbilicus and had removed it with tweezers. Five days later, the patient had fever, loss of appetite, fatigue, and redness around the umbilicus (Panel A). These symptoms abated after 4 days. Physical examination showed marked inguinal lymphadenopathy on both sides (Panel B). Treatment with oral ciprofloxacin was initiated for suspected ulceroglandular tularemia. Serologic testing supported the diagnosis; the *Francisella tularensis* antibody titer was 1:1280. Two weeks after the completion of treatment, there was a reduction in the lymphadenopathy. After an additional 2 weeks, the swelling had completely resolved. {from [N Engl J Med 2021; 384:1349; April 8, 2021](#)}



Summary slide

- A wide variety of zoonotic infections are caused by blood-borne pathogens.
- Many of these agents are transmitted by various insect or arthropod vectors, or certain animal reservoirs.
- Transient skin lesions are often associated with these infections with extracutaneous disease occurring via hematogenous spread of the microbial agent.



References

1. Murray PR, et al., *Medical Microbiology, 9th edition*: selected pages of chapters 20, 25, 29, 32, 34.

<https://www.clinicalkey.com/#!/browse/book/3-s2.0-C20180000924>

2. Tao, LE, V. Bhushan and M. Sochat. 2025. *First Aid for the USMLE Step 1*; McGraw Hill Publication.

Thank you

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>